

Smart Traffic Lights Using Local Artificial Intelligence

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Abstract—Smart traffic-light systems can improve traffic management by adapting signal timing according to real-time road conditions. Traditional traffic lights generally operate using pre-defined timing patterns and may not respond effectively to sudden changes in traffic density. This paper presents a conceptual local-AI-based traffic signal system that uses camera input and local processing to analyze traffic conditions and make signal-control decisions. Local processing can reduce dependence on continuous cloud communication and support faster decision making. The proposed system also considers emergency-vehicle detection as an important extension for creating a clear route through nearby intersections.

Index Terms—Artificial Intelligence, Smart Traffic Lights, Machine Vision, Traffic Signal Control, Edge Computing, Emergency Vehicle Detection

I. INTRODUCTION

Traffic congestion is a major challenge in modern urban transportation systems. Conventional traffic signals generally operate according to predefined timing schemes, which may not respond effectively to rapidly changing traffic conditions.

Artificial intelligence and machine learning techniques have therefore been investigated for adaptive traffic signal control. Deep reinforcement learning has been studied as an approach for making traffic signal control decisions in complex traffic environments [1].

Machine-vision-based systems can obtain traffic information from cameras and use vehicle-related information for adaptive signal control [2]. Reinforcement learning approaches such as IntelliLight have also explored dynamic traffic signal control using real traffic data [3].

The use of learning-based methods for coordinated traffic signal control has also been investigated through approaches such as PressLight, which considers traffic-network coordination [4].

In addition to general traffic optimization, emergency vehicles require rapid passage through congested intersections. Recent research has investigated deep learning-based emergency-vehicle detection combined with traffic-signal synchronization [5]. Other recent work has explored YOLOv8-based emergency-vehicle detection and traffic-signal prioritization [6].

The objective of this work is to study a local-AI-based traffic-light architecture in which road information is processed

near the traffic intersection, allowing signal decisions to be generated using current traffic conditions.

II. LITERATURE REVIEW

Previous research has investigated several approaches to intelligent traffic signal control.

Rasheed et al. presented a review of deep reinforcement learning methods applied to traffic signal control and discussed different architectures, simulation platforms, and research challenges [1].

Gorodokin et al. investigated adaptive traffic-light control using machine vision. Their approach used camera-based traffic information and vehicle-flow parameters to adjust traffic-light operation [2].

Wei et al. proposed IntelliLight, a reinforcement-learning approach for intelligent traffic-light control and evaluated the method using real traffic data obtained from surveillance cameras [3].

PressLight investigated reinforcement learning for coordinating traffic signals in an arterial network and considered traffic pressure in the signal-control process [4].

More recent research has focused on emergency-vehicle priority. Ibraheem et al. proposed a system combining deep-learning-based emergency-vehicle detection with traffic-signal synchronization [5]. Shyamala et al. investigated YOLOv8-based emergency-vehicle detection combined with graph neural network-based traffic-signal prioritization [6].

III. METHODOLOGY

The proposed system follows a local-AI architecture for traffic signal control. A camera captures traffic conditions at an intersection. The captured information is processed by a local computing unit to identify vehicles and estimate traffic conditions.

Based on the processed information, the controller determines which traffic direction requires priority. The corresponding traffic signal is then adjusted.

The local processing approach is intended to reduce the dependence on continuous communication with a remote cloud server. The system can also be extended to detect emergency vehicles and provide priority to them.

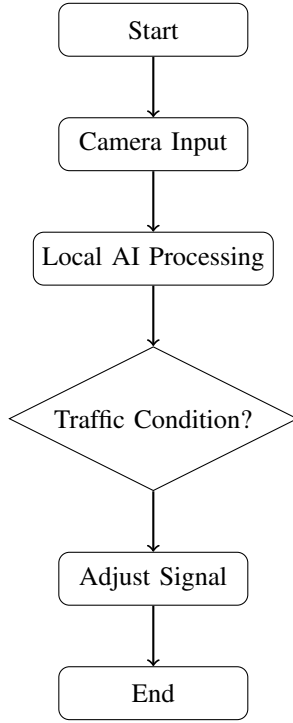


Fig. 1. System Flowchart

A. System Flowchart

IV. SYSTEM ARCHITECTURE

The proposed architecture consists of street cameras, local processing units, an optional cloud server, and traffic-light controllers.

The cameras collect information about vehicles and traffic conditions. The local AI unit processes this information and determines the appropriate signal action. Selected information may optionally be transmitted to a cloud server for storage and wider analysis.

V. RESULTS

The following table provides a sample comparison of different traffic-signal approaches.

TABLE I
COMPARISON OF TRAFFIC SIGNAL APPROACHES

Method	Decision Time	Internet Dependence
Fixed-Time	High	None
Cloud AI	Medium	High
Local AI	Low	Low

The values in Table I are conceptual comparisons for the proposed system and should not be presented as experimentally measured values.

VI. MATHEMATICAL MODEL

The traffic condition can be represented using the number of vehicles detected in each lane.

Let

$$D_i = \frac{N_i}{L_i} \quad (1)$$

where D_i represents the traffic density of lane i , N_i represents the number of detected vehicles, and L_i represents the considered road length.

The relative traffic density can be used to determine which traffic direction should receive signal priority.

A. Matrix Representation

The traffic information from different lanes can be represented using the following matrix:

$$A = \begin{bmatrix} D_1 & D_2 \\ D_3 & D_4 \end{bmatrix} \quad (2)$$

where D_1 , D_2 , D_3 , and D_4 represent the estimated traffic densities of four approaches to an intersection.

VII. ALGORITHM

Algorithm 1 Local AI Traffic Signal Control

- 1: Start
 - 2: Capture traffic information using camera
 - 3: Process the input using the local AI model
 - 4: Estimate vehicle density
 - 5: Check for emergency vehicles
 - 6: Determine signal priority
 - 7: Adjust traffic signal
 - 8: Repeat the process
 - 9: Stop
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VIII. CONCLUSION

This paper presents a conceptual local-AI-based approach for adaptive traffic signal control. The proposed system uses camera-based traffic information and local processing to make signal-control decisions according to current traffic conditions. Previous research demonstrates the use of reinforcement learning and machine vision for adaptive traffic signal control, while recent studies have also investigated emergency-vehicle detection and signal prioritization.

The proposed architecture can be further developed by implementing a real-time vehicle-detection model, testing the system using traffic datasets or simulation platforms, and evaluating its performance under different traffic conditions.

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